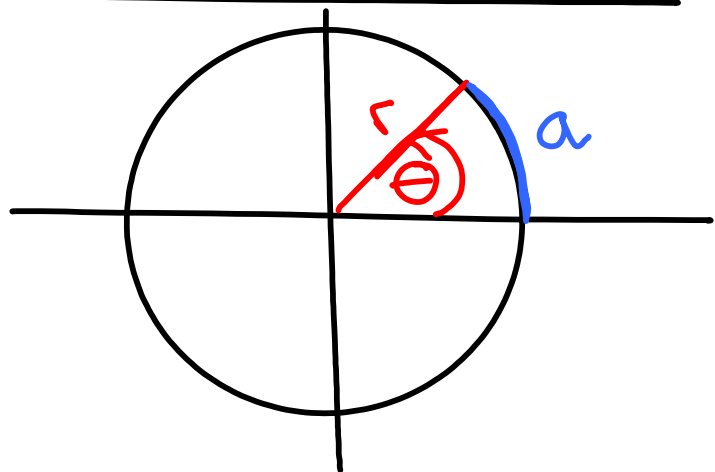


Radiaalmaat en Trigonometrische Functies

(Studiegids laatste bl.)

Definitie:



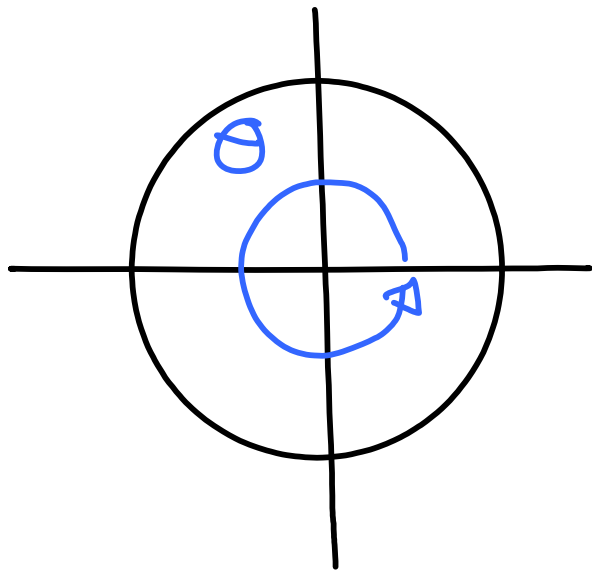
$$\pi \text{ rad} = 180^\circ$$

a is booglengte

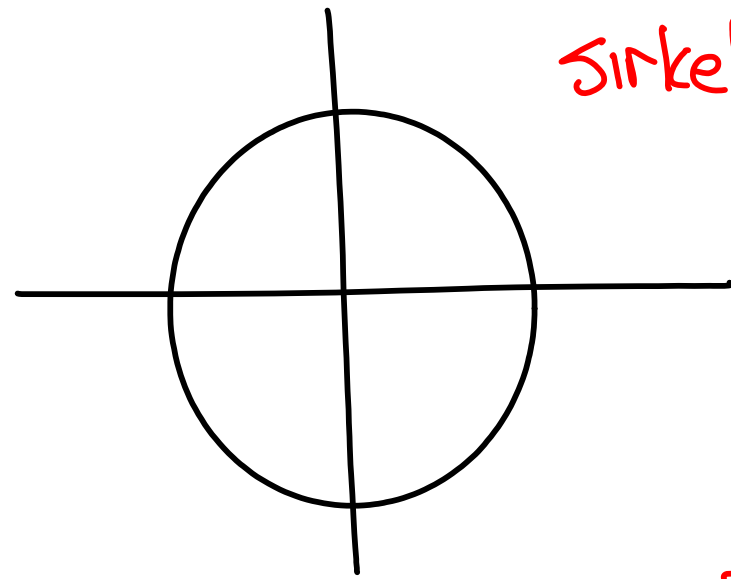
$$\theta = \frac{a}{r} \text{ radiale}$$

$$\pi = \text{pi} \approx 3.14 \text{ (reële getalle)}$$

Omrekening tussen graden en meting in radiale.



$$\Theta = 360^\circ$$



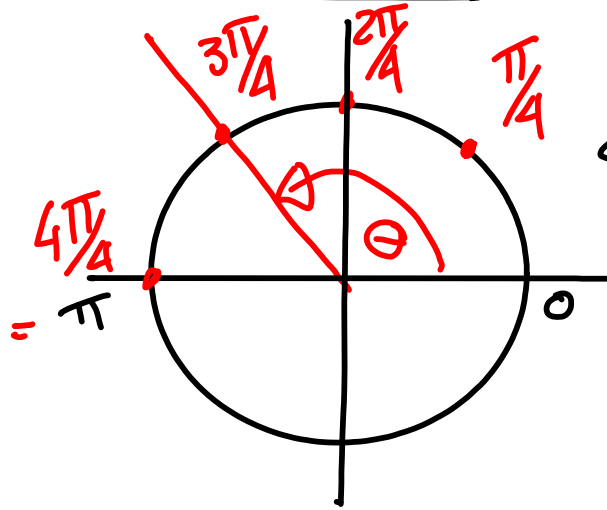
Strecke
= $2\pi r$

$$\Theta = \frac{s}{r} = \frac{2\pi r}{r} = 2\pi$$

360° is 2π radiale
 180° is π radiale

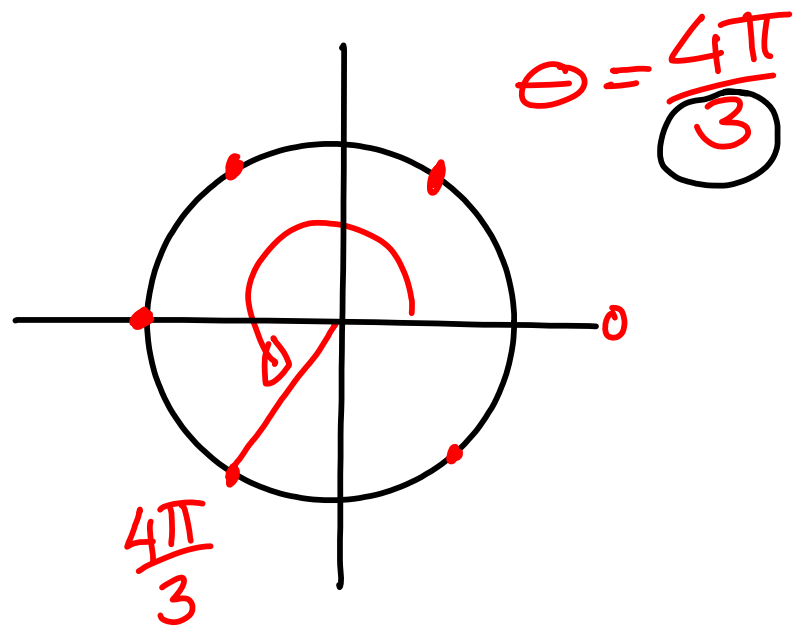
Grade	0°	30°	45°	60°	90°	180°	360°
Radiale	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	π	2π

Sketchs:



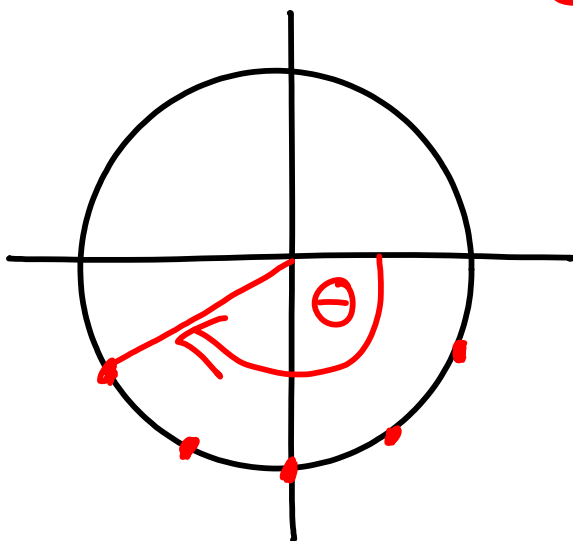
$$\theta = \frac{3\pi}{4}$$

(4)

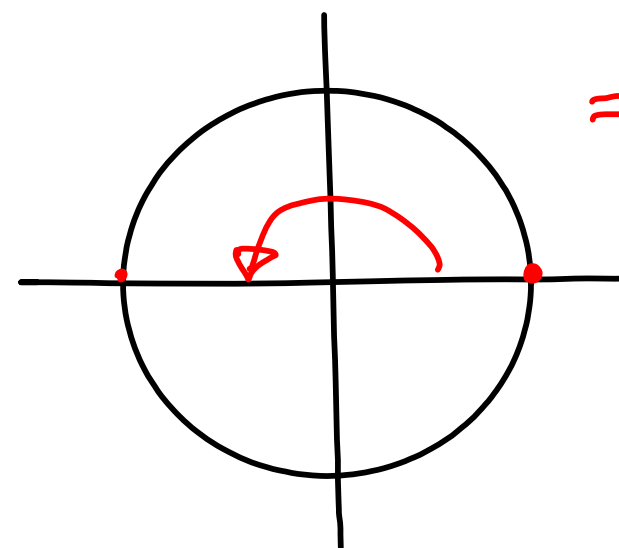


$$\theta = \frac{4\pi}{3}$$

(3)



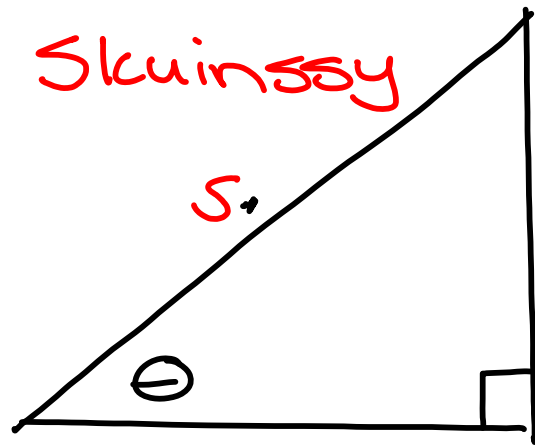
$$\theta = -\frac{5\pi}{6}$$



$$\theta = 21\pi$$

$$= 10 \times 2\pi + \pi$$

Definisie : 6 Trigonometrisiese Funksies



skuinssy

s

t

teenoorstaande sy

θ

a

aangrensende sy

$$\bullet \sin \theta = \frac{t}{s}$$

$$\bullet \cos \theta = \frac{a}{s}$$

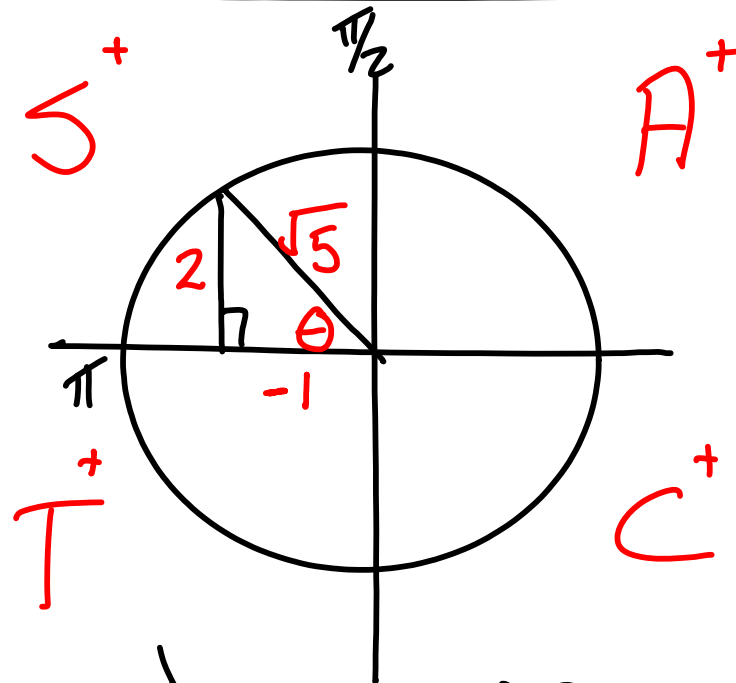
$$\bullet \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{t}{a}$$

$$\bullet \operatorname{cosec} \theta = \frac{s}{t}$$

$$\bullet \sec \theta = \frac{s}{a}$$

$$\bullet \cot \theta = \frac{a}{t}$$

Tekens van Trig Functies



Gegee:

• $\tan \theta = -\frac{2}{1}$,
 $\frac{\pi}{2} < \theta < \pi$

- Vind waarden van ander trigonometrische verhoudings.

$$\sin \theta = \frac{2}{\sqrt{5}}$$

$$\cos \theta = \frac{-1}{\sqrt{5}}$$

$$\operatorname{cosec} \theta = \frac{\sqrt{5}}{2}$$

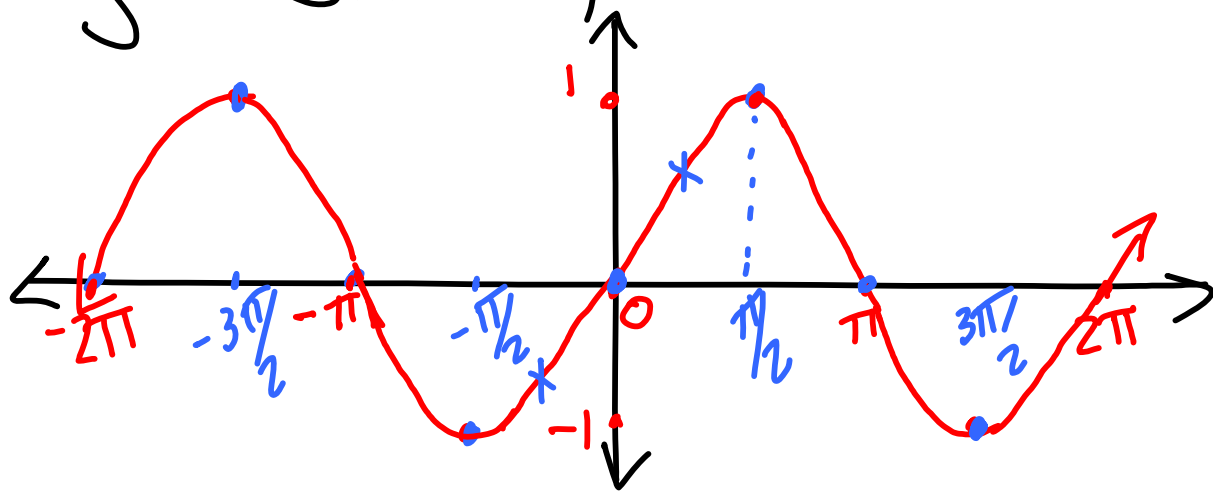
$$\sec \theta = -\frac{\sqrt{5}}{1}$$

$$\cot \theta = -\frac{1}{2}$$

Grafieken

A30 (Mod kan skets)

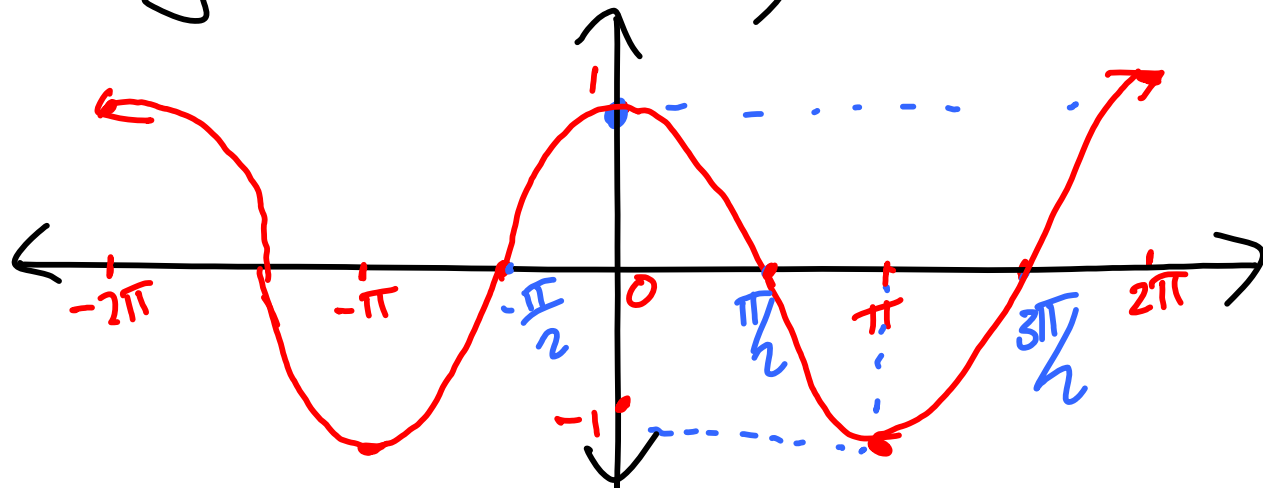
• $y = \sin x, x \in \mathbb{R}$



$$D_f = \mathbb{R}$$

$$W_f = \{y \mid -1 \leq y \leq 1\}$$

• $y = \cos x, x \in \mathbb{R}$



$$D_f = \mathbb{R}$$

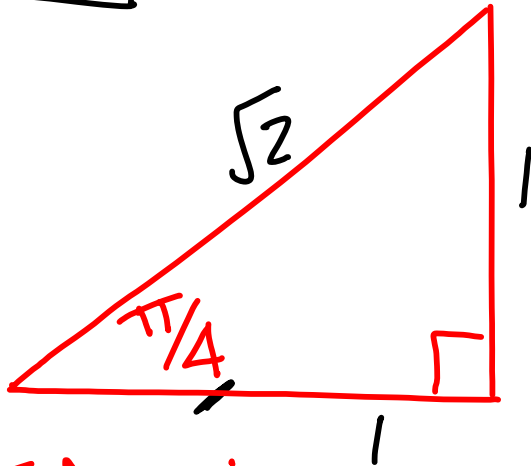
$$W_f = \{y \mid -1 \leq y \leq 1\}$$

Selfstudie : $y = \tan x$

Bepaal trigonometrische Verhoudings sonder sakrekenaar.

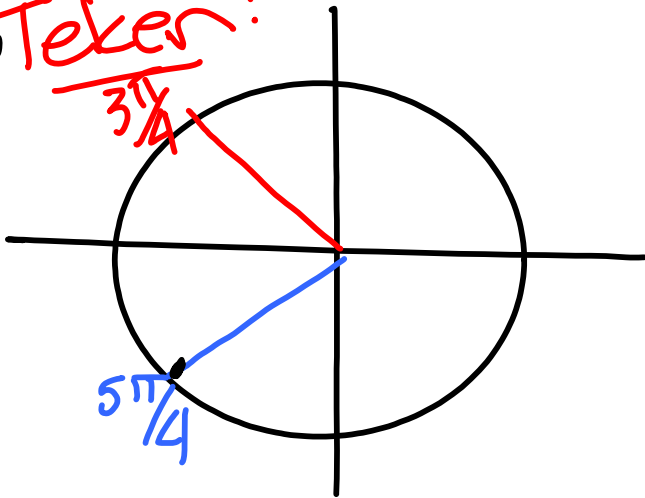
$\frac{\pi}{4}$

Verwysingshoek



$$\textcircled{\bullet} \operatorname{cosec} \frac{3\pi}{4} = +\sqrt{2}$$

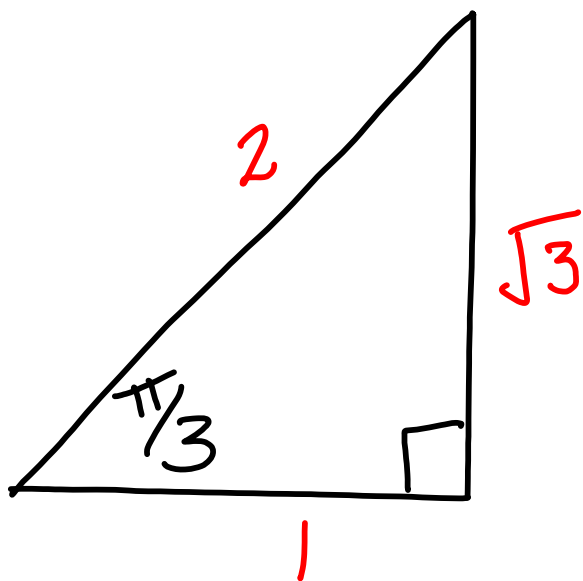
• Teker:
 $\frac{3\pi}{4}$



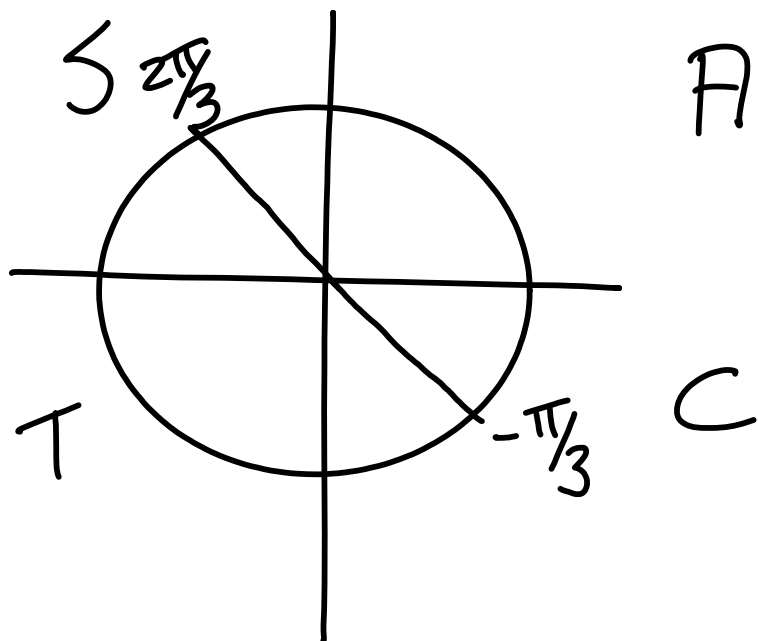
$$\textcircled{\bullet} \cos \frac{5\pi}{4} = -\frac{1}{\sqrt{2}}$$

$$\boxed{\frac{\pi}{3}}$$

verwysingshoek



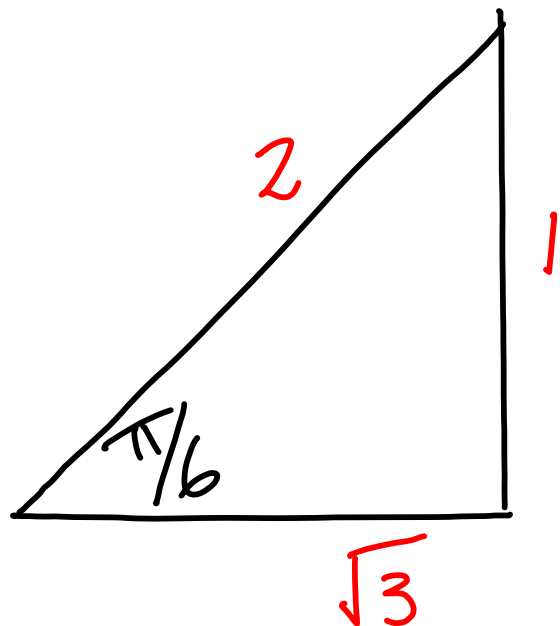
$$\bullet \tan\left(-\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{1}$$



$$\bullet \sec \frac{2\pi}{3} = -2$$

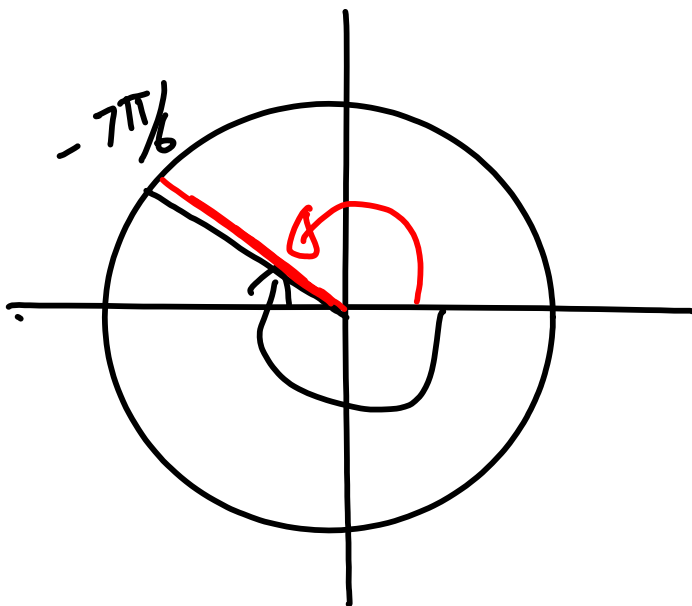
$\boxed{\frac{\pi}{6}}$

Verwysingshoek



$$\bullet \sin\left(-\frac{\pi}{6}\right) = +\frac{1}{2}$$

$$\bullet \cot\left(\frac{5\pi}{6}\right) = -\sqrt{3}$$



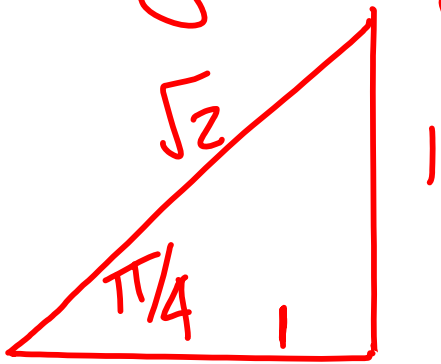
- $\sin(3\pi)$ $3\pi = 2\pi + \pi$
 $= \sin \pi$
 $= 0$

- $\cos(-\frac{9\pi}{2})$ $\frac{9\pi}{2} = 4\frac{1}{2}\pi$
 $= \cos(-\frac{\pi}{2})$ $= 4\pi + \frac{\pi}{2}$
 $= 0$

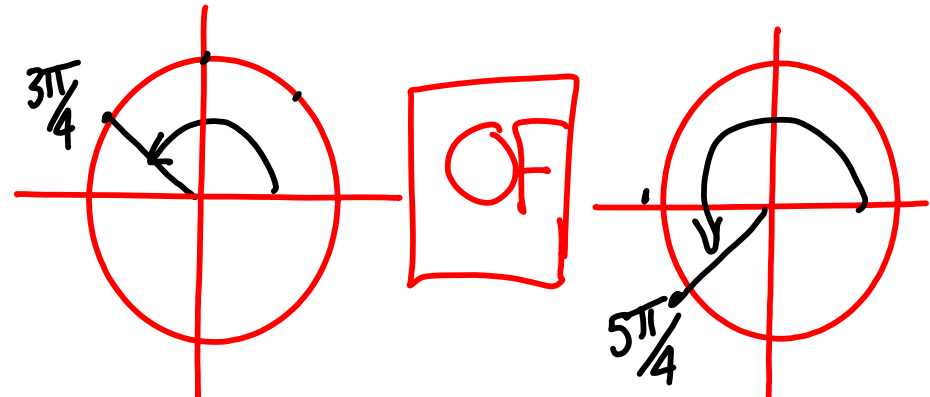
Oplos van Trigonometrie

Vergelykings

① $\cos 2x = -\frac{1}{\sqrt{2}}$
 Verwysingshoek



Teken



$$2x = \frac{3\pi}{4} + 2k\pi$$

$$\text{of } 2x = \frac{5\pi}{4} + 2k\pi$$

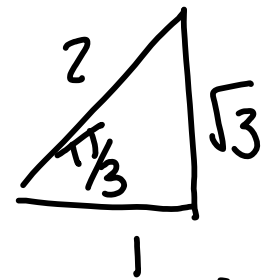
$$\Rightarrow x = \frac{3\pi}{8} + \cancel{2k\pi}$$

$$\text{of } x = \frac{5\pi}{8} + \cancel{2k\pi}$$

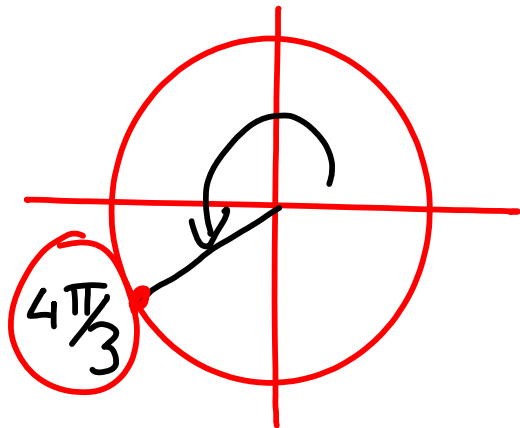
$$\text{waar } k \in \mathbb{Z}$$

$$\textcircled{2} \operatorname{cosec} x = -\frac{2}{\sqrt{3}}$$

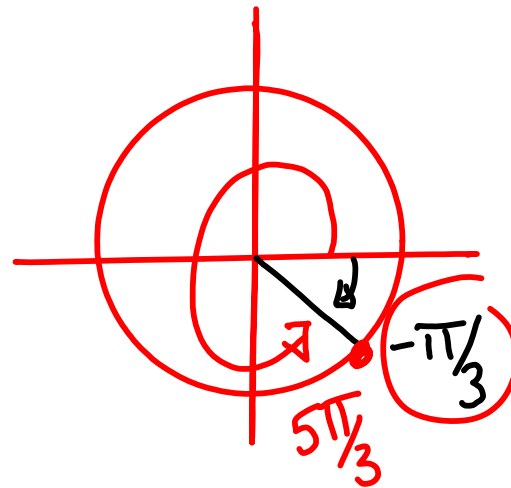
- Verwijnschoek: $\frac{\pi}{3}$
- Tekens:



Geen beperking!!



of



$$x = \frac{4\pi}{3} + 2k\pi \quad \text{of} \quad x = -\frac{\pi}{3} + 2k\pi \cdot \left. \begin{array}{l} \\ = \frac{5\pi}{3} + 2k\pi \cdot \end{array} \right\}$$

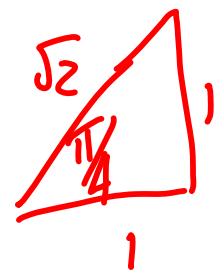
Maar wat van:

$$\operatorname{cosec} x = -\frac{2}{\sqrt{3}}, \quad x \in \underbrace{[0, 2\pi]}_{\text{Beperking}}$$

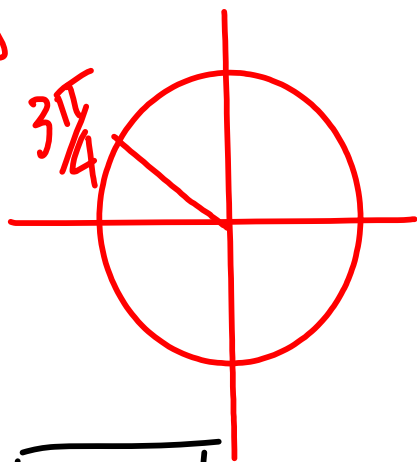
Dar: $x = \frac{4\pi}{3}$ of $x = \cancel{-\frac{\pi}{3}} \quad \frac{5\pi}{3}$

$$\textcircled{3} \quad \tan \frac{x}{2} = -1$$

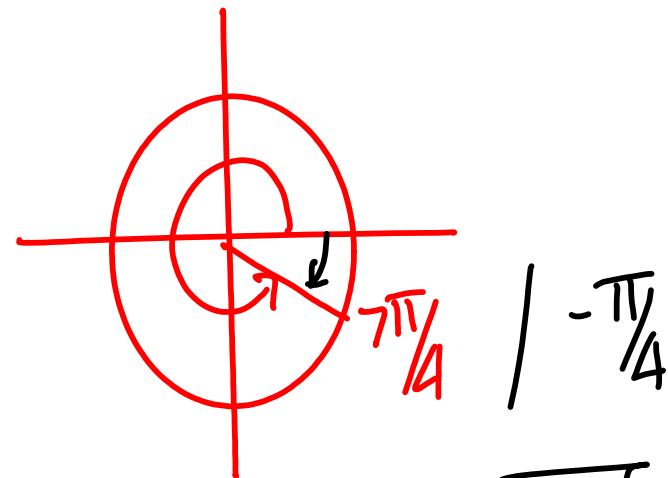
• Verwysingshoek: $\frac{\pi}{4}$



• Teken



of



$$\frac{x}{2} = \frac{3\pi}{4} + \boxed{k\pi}$$

$$\textcircled{2} \quad \frac{x}{2} = \frac{\pi}{4} + \boxed{k\pi}$$

$$\Rightarrow x = \frac{3\pi}{2} + 2k\pi$$

$$\textcircled{2} \quad x = \frac{\pi}{2} + 2k\pi \quad k \in \mathbb{Z}$$

$$\left(x = -\frac{\pi}{2} + 2k\pi \right)$$

*ken Identiteite : A28

$$\textcircled{4} \quad \sin 2x = \cos x \quad \text{op } [0, 2\pi]$$

$$\Rightarrow 2 \sin x \cos x = \cos x$$

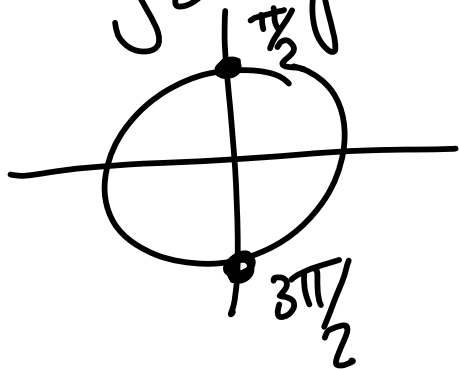
$$\Rightarrow 2 \sin x \cos x - \cos x = 0$$

$$\Rightarrow \cos x (2 \sin x - 1) = 0$$

$$\Rightarrow \cos x = 0 \quad \text{of} \quad 2 \sin x = 1$$

$$\sin x = \frac{1}{2}$$

Verwysingp: $\frac{\pi}{2}$



$$\therefore x = \frac{\pi}{2} \quad \text{of} \quad x = \frac{3\pi}{2}$$

of
=

of

Verwysingp: $\frac{\pi}{6}$



$$x = \frac{\pi}{6} \quad \text{of} \quad x = \frac{5\pi}{6}$$

Repeating

$$\textcircled{5} \quad 2 \cos^2 x - \sin x - 1 = 0$$
$$\Rightarrow 2(1 - \sin^2 x) - \sin x - 1 = 0$$

Gen. Bepending
 $+ 2k\pi$

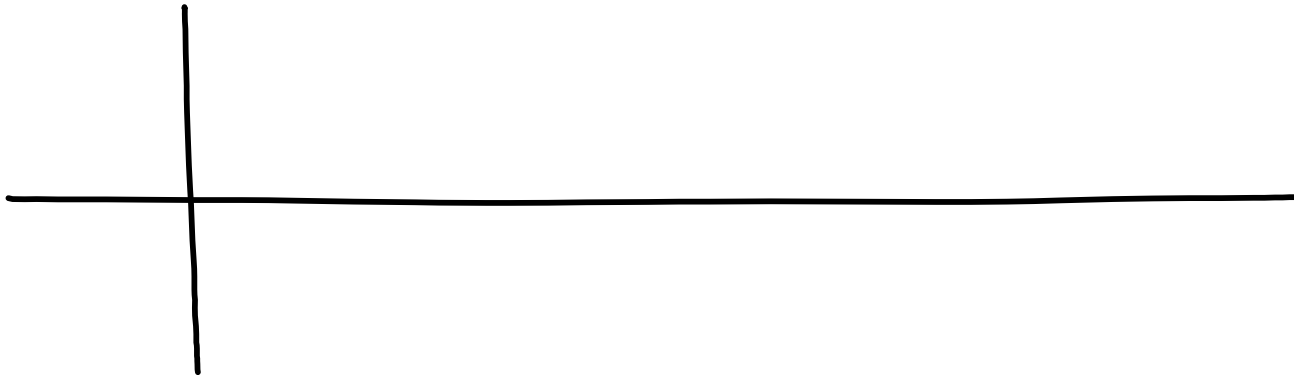
Trig Onge lykhede

* Moet grafiek gebruik

① $\sin x \leq \frac{1}{2}$, $x \in [0, 2\pi]$

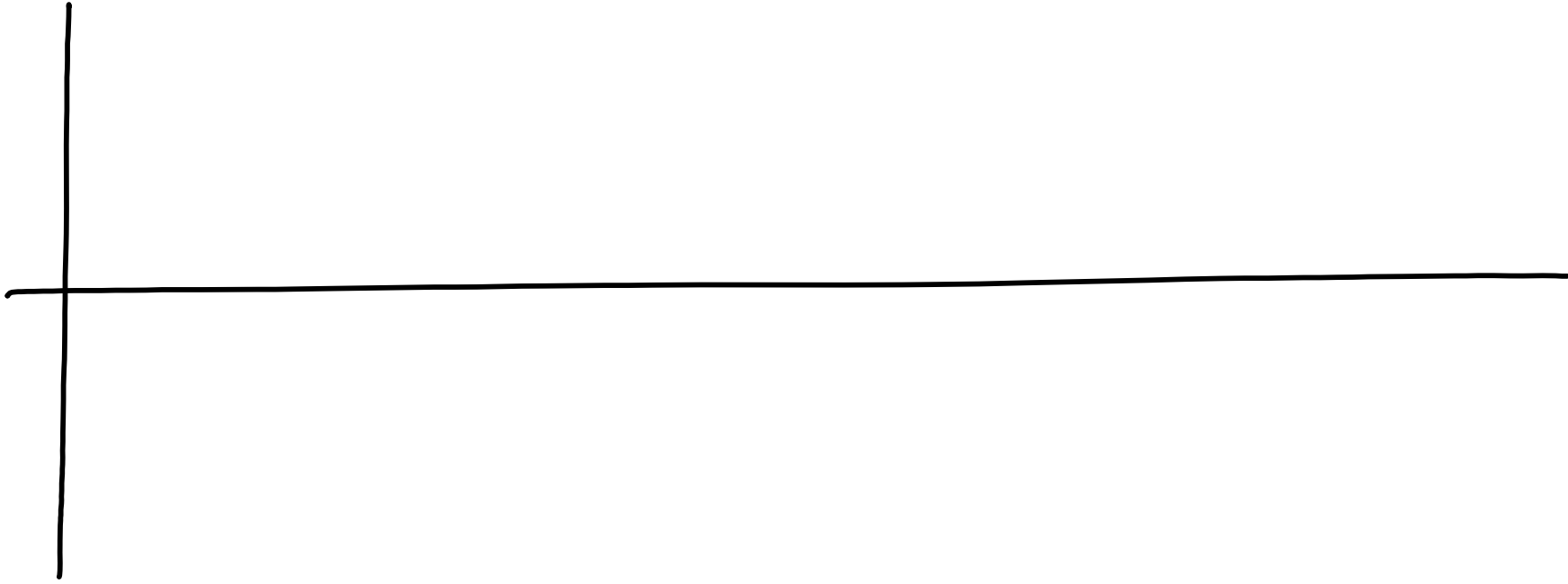
- $\sin x = \frac{1}{2}$

- $\sin x \leq \frac{1}{2}$



$$\textcircled{2} \quad 2\cos x - 1 > 0, \quad x \in [0, 4\pi]$$

Grafiek



Dus $\cos x > \frac{1}{2}$

$$\textcircled{3} \quad \tan x > -1$$

Probeer die folgende

1. $|\sin x| \leq \frac{1}{2}$, $x \in [-2\pi, 2\pi]$
2. $\cos x > \sin x$, $x \in [-2\pi, 2\pi]$
3. $|\cos x| \geq \frac{1}{2}$, $x \in [0, 2\pi]$
4. $0 \leq \tan x \leq 1$, $x \in [-\pi, 2\pi]$